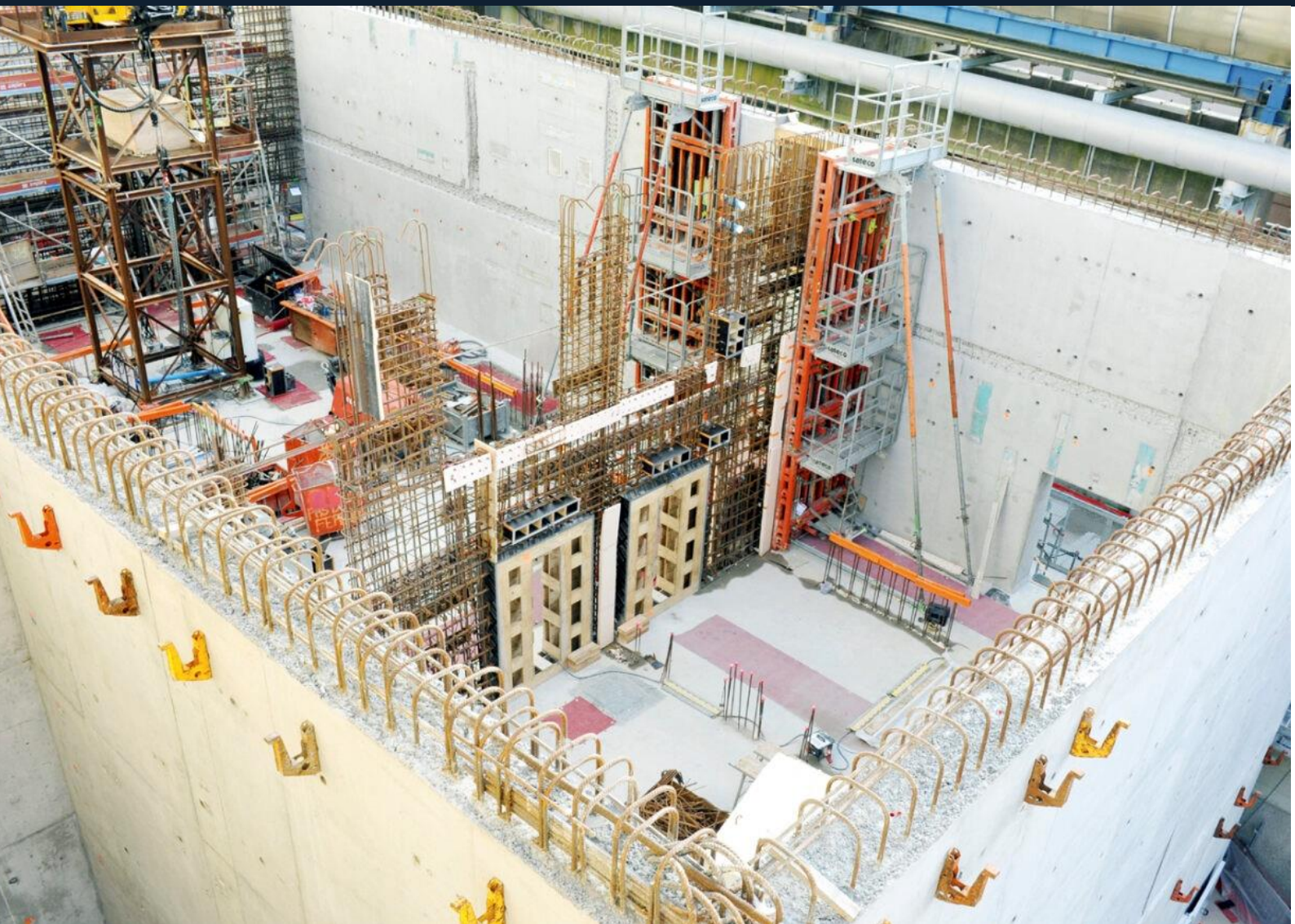


Health & Safety

Standard Stability



Reference

Company	Department	Document type	Sequence No.	Revision	Language
BYTP	H&S	REF	2213	D	EN

Version	Date	Written by	Checked by	Approved by	Reason for modification
A	01-11-2023	I.LELIEVRE	A.JANSSENS	L.KNOLL	First issue
B	21 May 2024	A.JANSSENS	I.LELIEVRE O. BRZEZINSKI	L. KNOLL	Modification of paragraph 3
C	14-10-2024	F. MARTIN	I.LELIEVRE O. BRZEZINSKI	L. KNOLL	Addition of the "Bolted Assemblies Recommendations" in section 3. Documents reference.
D	23-12-2025	I. MUCCIGNATO	F.MARTIN	L. KNOLL	Revised based on the "BYCN Guideline"

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1. Objective

The purpose of this document is to describe Bouygues Travaux Publics (BYTP) requirements in terms of stability.

This involves preventing the risks associated with the instability of the temporary works, in particular in accordance with the regulations, codes and standards in force in all the countries where BYTP operates.

2. Scope

This document applies to all companies working on sites (including service providers and subcontractors).

In the case of Joint ventures (JV), Consortiums or co-contracting operations, the project-specific procedure must include, as a minimum, the provisions set out in this document as well as any non-conflicting provisions of our partners. It is important to have all aspects adopted at the start of the project either prior to or as a minimum at Health and Safety kick-off meetings.

If any part of this document conflicts with a local requirement and is less stringent than that local regulation, then the corresponding local regulation(s) apply.

3. Reference documents

- BYTP-H&S-REF-2200 – Standard Lines of defence and Life Saving Rules
- BYTP-H&S-REF-2210 – Standard Lifting
- BYTP-H&S-REF-2226 – Standard Earthworks activities
- BYTP-H&S-REF-2227 – Standard Underground construction work
- BYTP-H&S-REF-2230 – Standard Work with concrete
- BYTP-H&S-INF-2166 – Standard practices for securing access to height and work at height
- BYTP-TEC-PRO-5637 – Managing and Controlling Temporary Works
- Policy regarding sensitive equipment BYTP and Subsidiaries
- BYCN Guidelines for the stability of works and storage – July 2025
- "Bolted Assemblies – Recommendations" of BYTP Equipment Department"
- MOSE+

4. Definitions and abbreviations

Term / Abbreviation	Description
TWC	Temporary Works Coordinator
Permanent works	Permanent works are parts of a construction project that will remain in place for a long time. This includes buildings and structures such as bridges, roads, retaining walls, etc. The erection of most types of permanent works will require the use of some form of temporary works.
Temporary work	Temporary works are elements that are erected to facilitate the construction of the permanent works, such as: • Shoring/Bracing • Scaffolding/work platforms designed to support personnel, equipment and materials

- Special equipment: Formwork of all kinds (climbing, sliding, tunnel, etc.)
- Protective devices against the risk of falling objects or equipment and the risk of impact
- Temporary bracing devices
- Retaining structures, shoring, excavation protection and dewatering
- Anchoring or foundations for lifting equipment

They will, most often, be removed once construction is completed.

The following are not considered temporary works:

- Handling equipment such as launching beams, cranes, gantries and lifting equipment, construction site lifts
- Temporary elements incorporated into permanent structures: precast slabs, reinforcement, etc.

Category of temporary work	Temporary works are categorised according to their level of complexity and the risks associated with their design, assembly/disassembly and use. Depending on local regulations, categories of temporary works are defined and associated management procedures are established. For example, in France, temporary works are divided into two categories according to their complexity and the level of risk to third parties or to the permanent structure. Category 1 structures are sensitive structures, in particular special equipment, as well as all structures that pose a risk to personnel, third parties and/or in an area open to the public. Category 2 structures are simple structures with a low level of risk (risk limited to site personnel).
TPS	Push-pull struts, also known as stabilisation struts

5. Roles and responsibilities

The roles and responsibilities associated with the design and monitoring of the execution of Temporary works are defined in detail in the Technical Department's document **BYTP Temporary Works procedure COP-TWM (BYTP-TEC-PRO-5637)**.

In particular, they are:

- The BYTP Lead Temporary Works Coordinator from the Technical Department, appointed by the BYTP Technical Director
- The Lead Temporary Works Coordinator representatives, defined for each operational area and appointed by the Technical Directors of BYTPRF and BYTP
- The Temporary Works Coordinators (TWC) appointed by the Project Director
- The construction teams

In addition, the Management team (project/entity) organises and ensures that inspection visits are carried out to check compliance with the provisions defined in this Standard.

6. Training, competencies and permits

6.1. Training

Some temporary works reception require certified training in accordance with the criteria defined by applicable local regulations. This applies in particular to:

- Fixed scaffolding
- Mobile scaffolding
- Access ladders
- Escalib tower
- Concrete ballast scaffolding
- Unloading bay
- Cantilevered working platform (SATECO P3D or PRM type)

6.2. Competencies

For each type of temporary work:

- The roles of design, inspection and control prior to use, loading or dismantling are carried out by competent persons who are clearly identified within the project organisation.
- Persons involved in the assembly, use, loading or dismantling of a temporary work have sufficient knowledge of the stability of the structure at each stage
- The acceptance of Temporary works is carried out by authorised personnel, subject to experience gained in the field

6.3. Permits

In certain working environments, the complexity of the activities and the regulations in force may require the establishment of work permits, signed and issued by designated and competent persons.

A single work permit may be issued for several identical operations covered by the same design, provided that each operation is subject to appropriate control before the start of the operation.

7. Design of temporary works

Each temporary work is subject to a design that includes:

- The dimensioning of the structures and technical means to be used at all stages (unloading, storage, assembly, lifting if necessary, temporary and permanent stability)
- The dimensioning of the ground load distribution according to the resistance of the ground
- The production of layout plans, execution plans and calculation notes
- The formalisation of operating procedures for its implementation
- Technical compliance of equipment contributing to its stability: study of locally approved materials that comply with the technical data to be taken into account (e.g. load transfer), local regulations and are in good condition

This design is carried out by a competent member of the technical department or by a specific design office.

The earthworks design takes the following factors into account:

- The potential risk of collapse of materials around the perimeter of the excavation
- The soil type assessment provided by a soil engineer
- Stability calculations indicating the type of protection systems (slope angles, shoring, benching, sheeting, etc.)
- The need for a well or pumping system for drainage
- Provisions for access and evacuation

Temporary works involving the use of standard equipment are covered by the technical manual or assembly plan for that equipment.

Temporary works involving scaffolding, formwork, etc. are not subject to specific design requirements, provided that:

- The site conditions do not conflict with the scope or limitations of use of this standard equipment.
- The configurations for their use are covered by the technical manual or the supplier's instructions.

When the work is subcontracted, the subcontractors provide studies justifying the stability conditions of the temporary works. The BYTP production team verifies that the calculation assumptions used are correct before BYTP employees, subcontractors or temporary workers have access to the temporary work or work equipment.

8. Management of temporary works

8.1. Organisation

- The project appoints one or more TWCs (depending on skills and specialties) from the Construction or Methods departments.
- The project establishes an internal Management of temporary works procedure and sets up an organisation to ensure that all temporary works on site are planned, designed, verified, assembled and dismantled safely, at the appropriate time, by trained and competent personnel.
- The project defines the checks to be carried out and the frequency of inspections to ensure that the measures planned to ensure the stability of the Temporary works are complied with.
- The project creates and maintains a register of Temporary works

Organisation with regard to subcontractors

For subcontracted work or equipment, a procedure is established by the project to define the respective roles and responsibilities between BYTP personnel and subcontractors (verification, acceptance, Go-NoGo) for each stage of the Temporary work (installation, modification, dismantling, etc.).

8.2. Inspections and checks

Inspections during assembly

- Targeted inspections are regularly organised on site to ensure that assembly is carried out in accordance with operating procedures.
- When temporary works are installed by more than one team, for example day and night shifts, the production manager organises a handover meeting between the teams to discuss the progress of the installation in relation to the design and any potential risks for the installation team taking over due to partially completed work.

Checks before use, loading or dismantling

- All equipment, materials and machinery are stabilised in accordance with technical provisions, operating procedures and validated methods: support, verticality, number and position of stability devices (anchors, bracing, etc.).
- Work areas to which people will have access are secured and safety features are in place, including collective protection (guardrails in place, hatches closed, etc.).
- No access to or loading of equipment may be authorised without confirmation that it has been assembled in accordance with the design.
- No dismantling of stability equipment without the formal agreement of the competent person.

Periodic inspections during use

- Regular inspections are organised to ensure that operating procedures are followed and that equipment is maintained in good condition and free from defects (cracks, corrosion, deformation). The frequency of these inspections depends on local regulations and the risk analysis carried out by the project.
- Specific inspections are carried out before work resumes following adverse weather conditions (wind, rain, frost, etc.) that may affect the condition and stability of structures and equipment.

Modifications or deviations

- During inspections and checks, any modifications or deviations from the design are identified and reported for consultation and compliance.
- Temporary works under construction or whose inspection has resulted in a "No Go" and which require modifications are made inaccessible (exclusion zone) and identified with clear signage.

Factory or supplier inspections

For prefabricated elements, inspections of manufacturing sites are organised based on the project's risk analysis. These visits ensure that our supplier have implemented quality controls on critical aspects of the manufacturing process (e.g. reinforcement, lifting inserts, stabilisation provisions, concrete quality, etc.).

9. Special temporary works

9.1. Working platforms at height

The **Standard practices for securing access to height and work at height (BYTP-H&S-INF-2166)** provides detailed specifications for the following equipment:

- Mobile elevating work platforms (MEWPs)
- Fixed and mobile scaffolding, shoring towers
- Individual mobile platform (light)
- Mast-mounted platforms
- Cantilevered work platforms
- Suspended platforms
- Temporary decking and walkways

9.2. Formwork

The **Standard Work with concrete (BYTP-H&S-REF-2230)** covers, among other things, stability issues associated with:

- Formwork, including formwork panels
- Horizontal formwork
- Shotcrete

The **Standard Underground construction work (BYTP-H&S-REF-2227)** covers, among other things, stability issues associated with:

- Vault formwork
- Temporary supports

9.3. Bracing pull/push props

Refer to the Company Standard Operating Procedures (MOSE+)

- The dimensions of the Bracing pull/push props and connection cap anchors depend on the height of the prefabricated elements and the wind zone. They are indicated on the layout plan produced by the Methods Department
- Stability devices may only be removed following approval by the TWC

Points of vigilance (always refer to the requirements in the method plans):

- As far as possible, put the bracing pull/push props at a man's height in order to avoid working at height / arms above the shoulders
- The end opposite the structure to be stabilised must be connected to a single supporting structure (ground, slab or ballast block). When using a slab or ballast block, each element must be dedicated to bracing pull/push props in order to avoid uncertainties regarding the load path
- There must be at least 2 bracing pull/push props per element
- It is prohibited to use hammer-in anchors to stabilise prefabricated parts

- Use exclusively of M16 socket on vertical surfaces or HUS screws on the ground may be used (HUS Evolution 3 screws may be reused if they are checked against the template, otherwise reuse is prohibited).
- "ARTEON" rods do not absorb shear forces. PROHIBITION to use them to anchor a screed

9.4. Liernes (Vault) and buttons

Refer to the Company Standard Operating Procedures (MOSE+)

The Liernes (Vault) and buttons make it possible to maintain a fixed gap between 2 vertical structures or supporting the weight of a horizontal structure.

The sizing and installation method statement of the Liernes (Vault) and buttons must be carried out by the Technical Department and TWC acceptance is mandatory.

Welding is carried out by qualified personnel (certified for the type of weld to be carried out) and are performed according to the defined welding parameters.

Installation of tie Lierne

- Install the tie bar supports (2 per lierne) or lierne support brackets.
- Provide anti-slip angles
- Favor the assembly of the liernes by jointing or by end plates
- Workstations allow operators to work safely

Installation of buttons

The spacing between buttons also takes into account the size of the formwork and other loads to be handled

To facilitate removal:

- Equip the props with lifting lugs
- Provide a support system on the lierne for all props
- Systematically install buttons by bolting them on one side and using a "retractable" device on the other (Ischebeck wedges, jacks, etc.)

9.5. Soils and embankments

The soil stability requirements for the use of lifting equipment (MEWPs, tower cranes, etc.) are covered in the Standard **Lifting (BYTP-H&S-REF-2210)**.

Slope stability requirements are covered in the Standard Earthwork (**BYTP-H&S-REF-2226**).